



Perimeter, Area & Algebraic Expressions

Course: Intermediate Algebra · **Unit 1:** Expressions · **Deck:**
U1L15_Perimeter,_Area_&_Algebraic_Expressions.pptx (22 slides)

 You do not need to know algebra to teach this lesson. Read what is written. Follow the slides. Check answers against the key. The learning expectations for today are the same as any other day — this plan gives you the tools to keep the lesson moving. The art of teaching stays with your teacher; the tools are right here.

QUICK START — BEFORE THE BELL (2 MINUTES)

- ☐ Open the slide deck: **U1L15_Perimeter,_Area_&_Algebraic_Expressions.pptx**
- ☐ Write the Bell Ringer on the board (see Slide 1 / Answer Key below)
- ☐ Test the projector
- ☐ Write your name on the board
- ☐ Find the answer key — it is printed inside this plan

MATERIALS FOR THIS LESSON

- ☐ Whiteboards, markers, erasers (1 set per student)
- ☐ Direct Instruction slide deck (Navy 1C2950 headings, Arial body)
- ☐ Tiered Independent Worksheet (T1/T2/T3 versions)
- ☐ Practice materials for: Large Group Practice
- ☐ Exit Ticket slips / LMS exit ticket
- ☐ Visual vocabulary anchor chart
- ☐ Theme visual: Generic Amusement Park

If anything is missing, check the teacher's desk or ask the front office.

PACING AT A GLANCE

Slide	By minute	Phase
1	0–4	Bell Ringer
7	15–16	End of Direct Instruction
19	42	End of Practice (worksheet + game/activity)
22	50	Exit Ticket & Wrap-Up

Pacing tip: bell ringer ~4 min · title/I-can/vocab ~4 min · collab ~4 min · worked example + guided practice ~7 min · culture + career ~6 min · GO/SET/EXTRA ~12 min · game/activity ~4 min · individual solve ~4 min · summary + exit ~4 min. Keep the timer visible; the deck drives the pace.

SUBSTITUTE SCRIPT (SLIDE BY SLIDE)

Slide 1 — Bell Ringer

On screen: WHITEBOARDS READY! — Tiered prompts (T1 one-step, T2 two-step, T3 challenge) shown on the slide.

Say: “Whiteboards ready — three tiers, four minutes. Choose your tier, work silently, show your steps, and hold your board up when the timer hits zero.”

Do: Walk the room as they work; cue “3... 2... 1 — boards up” and glance for accuracy.

Answer key: Answers are revealed on the slide after the timer. Bridge to today’s topic: Perimeter, Area & Algebraic Expressions.

Slide 2 — Title

On screen: INTERMEDIATE ALGEBRA — UNIT 1: LESSON 15 — PERIMETER, AREA & ALGEBRAIC EXPRESSIONS. Minnesota standards: 9.3.5.7, 9.3.5.8. MERIT EDUCATION GROUP.

Say: “Welcome. Today’s lesson is Unit 1, Lesson 15: Perimeter, Area & Algebraic Expressions. Everything you need is on these slides — follow along and write when I ask.”

Do: Read the lesson title aloud. Pause for students to find a pencil.

Slide 3 — I Can Statements

On screen: I CAN STATEMENTS — Write expressions for perimeter and area; Use formulas for rectangles and composite shapes; Simplify geometric expressions.

Say: “Here’s what we’ll be able to do by the end of the lesson. Read them silently; we’ll come back to these at the end.”

Do: Give 15 seconds, then continue.

Slide 4 — Vocabulary

On screen: VOCABULARY — Perimeter — the distance around a shape; Area — the space inside a shape; Expression — write the formula with variables.

Say: “Quick vocabulary check. These are the words for today — I’ll read them, you repeat. [Read each term aloud.] We’ll use them all lesson.”

Do: Point to each term as you read it.

Slide 5 — Collaboration: Collaborative Whiteboard

On screen: ENGAGEMENT: COLLABORATIVE WHITEBOARD — prompt on the slide, tied to Perimeter, Area & Algebraic Expressions.

Say: “Collaborative Whiteboard. [Read the prompt aloud.] THINK silently for 30 seconds, then work with your partner or group, then we’ll share out.”

Do: Time 30s think, 45s pair, then take 2–3 volunteers.

Slide 6 — Worked Example (I Do)

On screen: WORKED EXAMPLE (I DO) — Step-by-step: A rectangle’s length is 3 more than its width w . Write an expression for the perimeter $\rightarrow 4w + 6$.

Say: “A worked example — watch how it’s done step by step. The slide shows every step; I’ll read them as they appear.”

Do: Read each step aloud as it is revealed. Answer questions by pointing back to the slide.

Slide 7 — Guided Practice (We Do)

On screen: GUIDED PRACTICE (WE DO) — Together: A square has side s . Write expressions for area and perimeter. A triangle has sides x , $2x$, and $x + 5$. Write its perimeter $\rightarrow A = s$ squared, $P = 4s$; $4x + 5$.

Say: “We do this one together. I’ll ask, you answer — then we check the slide.”

Do: Work step by step; call on students for the next step before revealing it.

Slide 8 — Culture

On screen: CULTURE: Sami (Sápmi / Northern Europe). The Sámi are the Indigenous people of Sápmi, spanning northern Norway, Sweden, Finland, and Russia. Sami culture centers on reindeer herding and duodji (handicraft).

Say: “A cultural connection. [Read the caption aloud.] Notice how math shows up in everyday life and community traditions.”

Do: If the slide shows an image only, describe what students see; no math required.

Slide 9 — Culture Word Problem

On screen: CULTURE WORD PROBLEM — A Sámi garment maker decorates a rectangular luhkka (reindeer herder) cloak. The length is $3x + 2$ and the width is $2x - 1$, all in units. Write an expression for the perimeter of the rectangle.

Say: “Here’s the cultural word problem. Read it to yourself — we’ll solve it together. Use the reading strategy on the slide: circle, underline, box, write.”

Do: Give 30 seconds to read; walk through the strategy, then the solution.

Slide 10 — Career

On screen: CAREER: PUBLIC WORKS INSPECTOR — City of Duluth, Duluth, MN. Entry wage: \$80,142/year (MN DEED OES, 25th percentile). Median wage: \$99,174/year.

Say: “A real career uses this math every day. [Read the career spotlight aloud.] The math we’re learning today is the math people use at work.”

Do: Read confidently; the salary and role are on the slide.

Slide 11 — Career Word Problem

On screen: CAREER WORD PROBLEM — A window energy rater at Andersen measures the length l and width w of each window. Write an expression for the perimeter and evaluate when $l = 3$ and $w = 2$.

Say: “Here’s the career problem. Read it to yourself — we’ll solve it together. Use the reading strategy on the slide.”

Do: Give 30 seconds; walk the strategy; then the solution.

Answer key: Answer is revealed on the slide (solution mirrors today’s practice: $4w + 6$).

Slide 12 — GO Practice (Your Turn)

On screen: GO PRACTICE: YOUR TURN — A rectangle’s length is 3 more than its width w . Write an expression for the perimeter

Say: “Your turn — a problem just like the one we did. Work it on your whiteboard (or paper). Two minutes.”

Do: Circulate. Cue boards up at 2 minutes.

Answer key: $4w + 6$

Slide 13 — GO Answer

On screen: GO ANSWER — reveal: $4w + 6$.

Say: “Let’s check it. The slide walks through the solution step by step — I’ll read each step. Check your work against it; fix anything you missed.”

Do: Read the solution steps aloud.

Slide 14 — SET Practice

On screen: SET PRACTICE — two problems: A square has side s . Write expressions for area and perimeter. A triangle has sides x , $2x$, and $x + 5$. Write its perimeter.

Say: “Two practice problems to deepen your understanding. Work them on paper or whiteboard. You have 5 minutes; I’ll be walking around.”

Do: Circulate and support. Cue time at 3 and 1 minutes.

Slide 15 — SET Answer

On screen: SET ANSWER — reveal: $A = s$ squared, $P = 4s$; $4x + 5$.

Say: “Let’s check. Read the steps on the slide; self-correct with a pen.”

Do: Read the steps aloud; answer questions by re-reading the relevant step.

Slide 16 — EXTRA Practice Challenge

On screen: EXTRA PRACTICE CHALLENGE — A rectangle has area 48 and width 6. Write and solve an equation for the length l

Say: “Challenge problem — for anyone who wants a stretch. Try it on your own paper. We’ll go over it in a moment.”

Do: Encourage all to attempt; it is optional.

Slide 17 — EXTRA Answer

On screen: EXTRA ANSWER — reveal: $6l = 48$, $l = 8$.

Say: “Let’s check the challenge. [Read the steps.] Great work.”

Do: Read the solution steps aloud; celebrate attempts.

Slide 18 — Large Group Practice

On screen: LARGE GROUP PRACTICE — directions on the slide.

Say: “Large Group Practice. [Read the directions aloud.] We’ll check together at the end.”

Do: Work the first problem together, step by step. Call on students to supply each next step. Students finish the remaining problems in pairs. Reveal answers and have students self-check.

Slide 19 — Individual Solve: Tiered Worksheet

On screen: INDIVIDUAL SOLVE: TIERED WORKSHEET — TIER 1 Build Confidence (6 questions), TIER 2 Apply Your Skills (6 questions), TIER 3 Challenge Yourself (6 questions). PROBLEMS: TIERED WORKSHEET — PERIMETER, AREA & ALGEBRAIC EXPRESSIONS + IN YOUR LMS. SHOW YOUR WORK.

Say: “Independent work now — the tiered worksheet. Pick your tier, show your work, and submit in the LMS when you finish.”

Do: Circulate; support tier choices; collect or confirm LMS submission.

Slide 20 — Summary

On screen: SUMMARY — RECAP: key ideas from Perimeter, Area & Algebraic Expressions. REVIEW: the skills practiced today. [Slide lists the takeaways.]

Say: “Three big takeaways from today. [Read them aloud.] This is what you should remember tomorrow.”

Do: Read each takeaway; pause between them.

IF A STUDENT ASKS...

“Why are we doing this?” — “This is today’s learning expectation for Perimeter, Area & Algebraic Expressions. It

builds toward what’s next in the unit, and your teacher will pick up right here

tomorrow.”

Redirects (from the lesson plan):

Misconception	Redirect Question	Corrective Strategy
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| Confuses perimeter with area | “Is the question about the distance around or the space inside?” | Draw the shape and label what is asked. |

| Adds length and width instead of doubling | “How many lengths and widths are in a rectangle’s perimeter?” | Write $2l + 2w$ and show each step. |

| Forgets units | “What units does the problem use?” | Carry units through every calculation. |

IF A STUDENT FINISHES EARLY

- Students who finish early do the **Challenge** side of the worksheet — do not invent extra work.
- If the whole class is ahead: re-read the Summary slide and ask students to explain one takeaway in their own words.
- **Do not** move ahead in the curriculum.

IF THE PROJECTOR DIES

Write the Bell Ringer problems on the board (from the Answer Key).

Read the worksheet directions aloud; students work in pairs.

Use the printed answer key to check as a class.

The lesson still runs — the script lives in this plan, not the projector.

WITH A PARAPROFESSIONAL OR CO-TEACHER

If an adult is in the room supporting students, they know the students best — **follow their lead** on individual support. Check the roster for accommodations; extended time on the Exit Ticket is fine if indicated.

SUB REPORT (LEAVE FOR THE TEACHER)

What did we complete? (bell ringer / DI / practice / game / exit ticket)

Who struggled or needed extra support?

Behavior notes (positive first, then concerns)

Anything tech-related that went wrong

TEACHER'S OWN NOTES

(The regular teacher may add specifics here — their timer, signals, roster quirks.)

Generated by the MEG Substitute Plan pipeline from the Lesson Plan + Teacher Edition — synced to the published slide decks (2026-08-19).